BARAKA PANDE BERENGE

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PROFILE

Computing & IT undergraduate (The Open University, expected 2028) with an Advanced Diploma in (Software Development, 95/100). I build and run self-hosted Linux, Proxmox VE and Kubernetes (k3s) infrastructure, plus AWS serverless environments provisioned with Terraform. Seeking a (Junior) cloud, systems or infrastructure engineering role.

CORE SKILLS

- **Infrastructure & Systems:** Proxmox VE | Linux (Ubuntu/Debian) | Docker | Kubernetes (k3s) | Nginx | Tailscale | Cloudflare Tunnel | ufw | Portainer | Uptime Kuma
- **Cloud & IaC:** AWS (S3, Lambda, API Gateway, DynamoDB) | Terraform
- **Networking:** TCP/IP | DNS | DHCP | VLANs | NAT | Subnetting |Routing
- **Development & Data:** Python (Flask) | REST APIs | WebSockets (Socket.IO) | PostgreSQL/PostGIS | Git | Bash | JavaScript | HTML/CSS
- **Automation:** GitHub Actions

FEATURED PROJECTS

Sentinel – Self-Hosted Device Tracking & Telemetry Platform (sentinel.barakaberenge.com)

- Built a personal device-tracking vault (Python, Flask, PostgreSQL, Socket.IO) with live location, history, battery monitoring, and offline and low-battery alerts via Discord webhooks.
- Added an opt-in Care Circle with isolated real-time rooms and a 4-stage SOS workflow (trigger, acknowledge, locate, resolve) with an audit log.
- Migrated from Docker Compose to a 3-node k3s cluster (Ubuntu 24.04 VMs on Proxmox VE) with local-path persistent storage and a Cloudflare Tunnel, with zero inbound web ports.
- Fixed an IDOR in the SOS API; added Flask-WTF CSRF protection, output escaping against XSS, hardened session cookies and 10-character random invite codes.

AWS Serverless Cloud Resume with Terraform & GitHub Actions (barakaberenge.com)

- Provisioned a serverless visitor counter backend on AWS (eu-north-1) using Terraform, configuring an S3 bucket with public access blocks, a DynamoDB on-demand table, a Python 3.12 Lambda function, least-privilege IAM roles, and an API Gateway v2 HTTP endpoint.
- Configured a GitHub Actions workflow to automate testing and Terraform deployment upon code push; decommissioned the environment cleanly at the end of the AWS free tier.

Self-Hosted Homelab Infrastructure

- Provisioned and administered a Proxmox VE hypervisor hosting Ubuntu VMs, Docker containers, Portainer, and Uptime Kuma monitoring, secured with ufw firewall rules and a 5-device Tailscale mesh network.
- Deployed a portfolio web application via Nginx and Gunicorn over Cloudflare Tunnel, resolving intermittent HTTP 502 gateway errors by forcing cloudflared from QUIC/UDP to HTTP/2 to eliminate NAT state collisions.

WORK EXPERIENCE

- IT Intern – Onda Informatica | Nov 2023 – June 2024
  Operated and monitored a production tracking system for hot steel blooms exiting the furnace, collaborating within a software development team to track system data flow during an ITS Academy internship.
- Reception & Administration – UIL Pineto | Nov 2021 – June 2022
  Managed administrative data systems and Client Inquiries to support daily branch operations.

EDUCATION

- BSc (Hons) Computing & IT – The Open University (Online) | 2025 – Present
- Advanced Diploma in Software Development – ITS Academy Lanciano | 2022 – 2024 (Final Grade: 95/100)
- High School Diploma – Nairobi | 2018 – 2022

TRAINING & CERTIFICATIONS

- **Cisco Networking Academy:** Network Technician Career Path (Completed 18 Jun 2026)
- **Cisco Networking Academy:** Linux Unhatched (Completed 18 Jun 2026)

LANGUAGES

- English – Fluent
- Italian – Fluent B2